DATA ANALYSIS PROJECT REPORT

Project Title	**Retail Sales and Profit Performance Analysis Using Python Data Analytics**
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Course / Section	B
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Date of Submission	03-05-2026

1. Executive Summary

This project analyzes a retail sales dataset using Pandas, NumPy, and Matplotlib to identify business trends, profit patterns, and customer purchasing behavior. The dataset contains over 9,000 records and multiple business-related attributes such as sales, profit, customer segment, shipping mode, and product category. The project aimed to investigate which regions and categories generate the highest profit, identify sales trends over time, and determine whether a relationship exists between sales and profit.

Several preprocessing steps were performed including duplicate removal, date conversion, and missing value handling. Feature engineering was applied by creating new columns such as profit margin and order month. NumPy was used to calculate z-scores for outlier detection and profit margin calculations.

The analysis revealed that the West region produced the highest sales, technology products generated the highest profit, and strong positive relationships existed between sales and profit. However, several orders produced heavy losses due to large discounts.

2. Introduction and Project Objectives

Item	Your Content
Dataset name	Sample Superstore Dataset
Dataset source	Dataset Link: https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Main project goal	Analyze sales and profit trends to identify business insights
Research Question 1	I will investigate which region generates the highest sales and profit?
Research Question 2	I will investigate which product category is the most profitable?
Research Question 3 (optional)	I will investigate if there is a relationship between sales and profit?

2. Dataset Description

The dataset contains customer order information from a retail superstore business.

Each row represents one customer transaction.

3.1 Basic Dataset Information

Property	Value
Number of rows	9994
Number of columns	21
Data types present	Numerical, Categorical, Date-time
Target or key variable(s)	Sales, Profit, Region, Category
Potential issues observed at first glance	Missing values and duplicate rows

3.2 Initial Observations

During initial inspection, the dataset showed a mixture of numerical and categorical variables. Date columns needed conversion into proper datetime format. Some duplicate records were identified. Several sales values were extremely high compared to average values, suggesting possible outliers.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification
Duplicate records	Removed duplicates	Prevent repeated analysis
Date stored as text	Converted to datetime	Required for trend analysis
Missing postal codes	Filled using median	Avoid null errors
Inconsistent formatting	Renamed columns	Improve readability

The cleaning process improved dataset quality and prepared the data for analysis and visualization.

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
Order Month	Extracted from Order Date	Monthly trend analysis
Profit Margin	Profit / Sales × 100	Business performance analysis
Sales Category	np.where() condition	Classify high and low sales
Order Year	Extracted from date	Year-wise comparison

These features made the analysis more meaningful and easier to interpret.

6. Analysis Methodology

- Pandas was mainly used for handling and analyzing the dataset efficiently. It helped in loading the CSV file, cleaning missing or duplicate values, filtering specific records, grouping data based on categories, and performing aggregation operations such as calculating averages, totals, and counts. These functionalities made the data preprocessing and analysis process faster and more organized.
- NumPy was used for performing numerical and mathematical operations on the dataset. It assisted in calculations such as z-score computation for outlier detection, conditional classification using numerical conditions, and other custom mathematical computations. NumPy also improved performance when working with large numerical datasets because of its optimized array-processing capabilities.
- Matplotlib was used for creating different types of charts and visual representations of the analyzed data. Using graphs such as bar charts, line charts, histograms, and pie charts, it helped present business insights in a more understandable and visually appealing way. These visualizations made it easier to identify patterns, trends, and comparisons within the dataset.

7. Analysis and Visualizations

7.1 Research Question 1

I will investigate which region generates the highest sales and profit?

To answer this question, the dataset was grouped by region using Pandas groupby() functions. Total sales and total profit were calculated for each region to compare business performance across geographical areas.

The analysis showed clear differences between regions in terms of revenue generation and profitability.

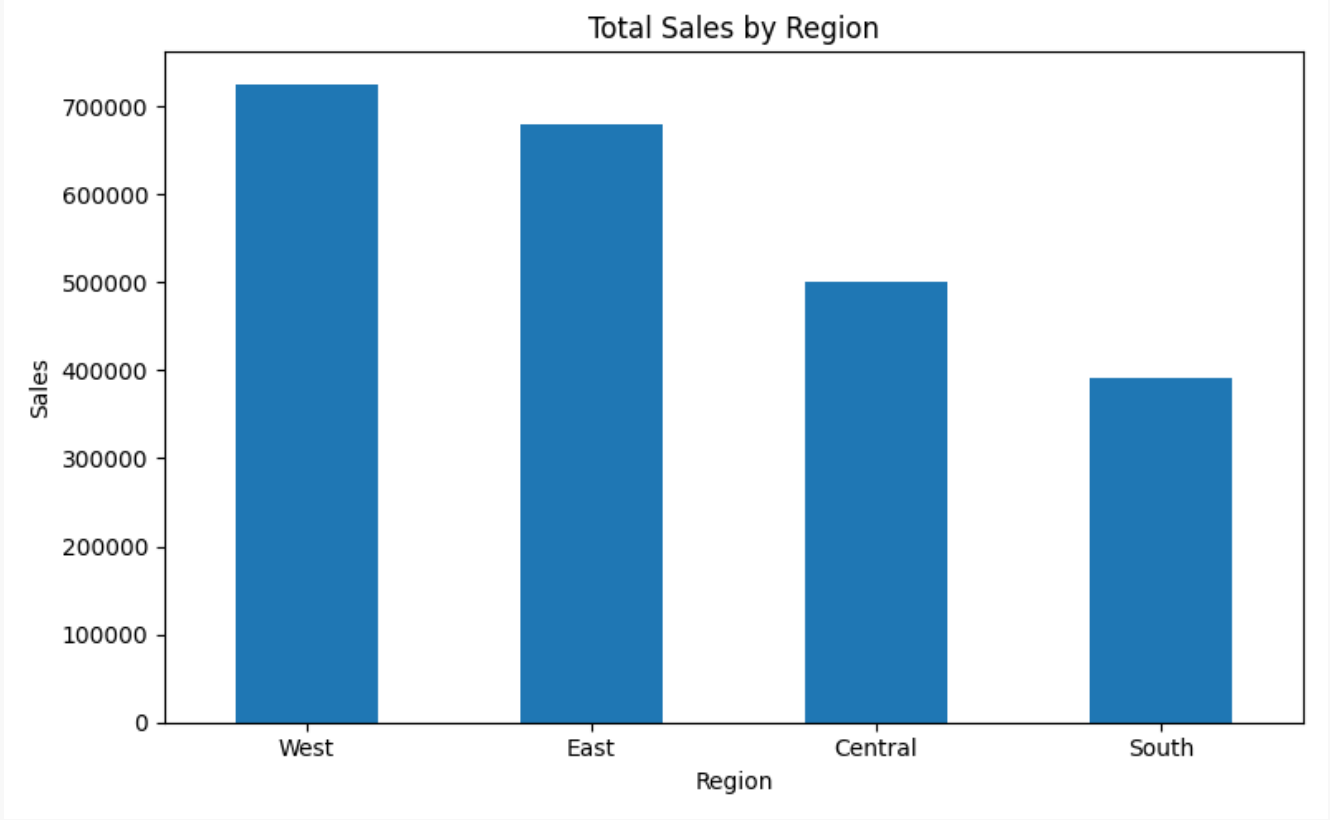

Figure 1. The figure shows the total sales generated by each business region. The West region produced the highest sales, while the South region generated comparatively lower sales revenue.

Interpretation: The West region contributed the largest share of total sales, indicating strong customer demand and market performance in that area. The East region also performed well but remained below the West region. The South region showed relatively weaker performance, which may suggest lower market penetration or fewer customer transactions. These results can help businesses prioritize investment and marketing strategies in high-performing regions.

7.2 Research Question 2

I will investigate which product category is the most profitable?

The dataset was grouped by product category to calculate total profit earned from each category. Category-wise aggregation helped compare business profitability between Furniture, Office Supplies, and Technology products.

The analysis identified significant differences in profitability among product categories.

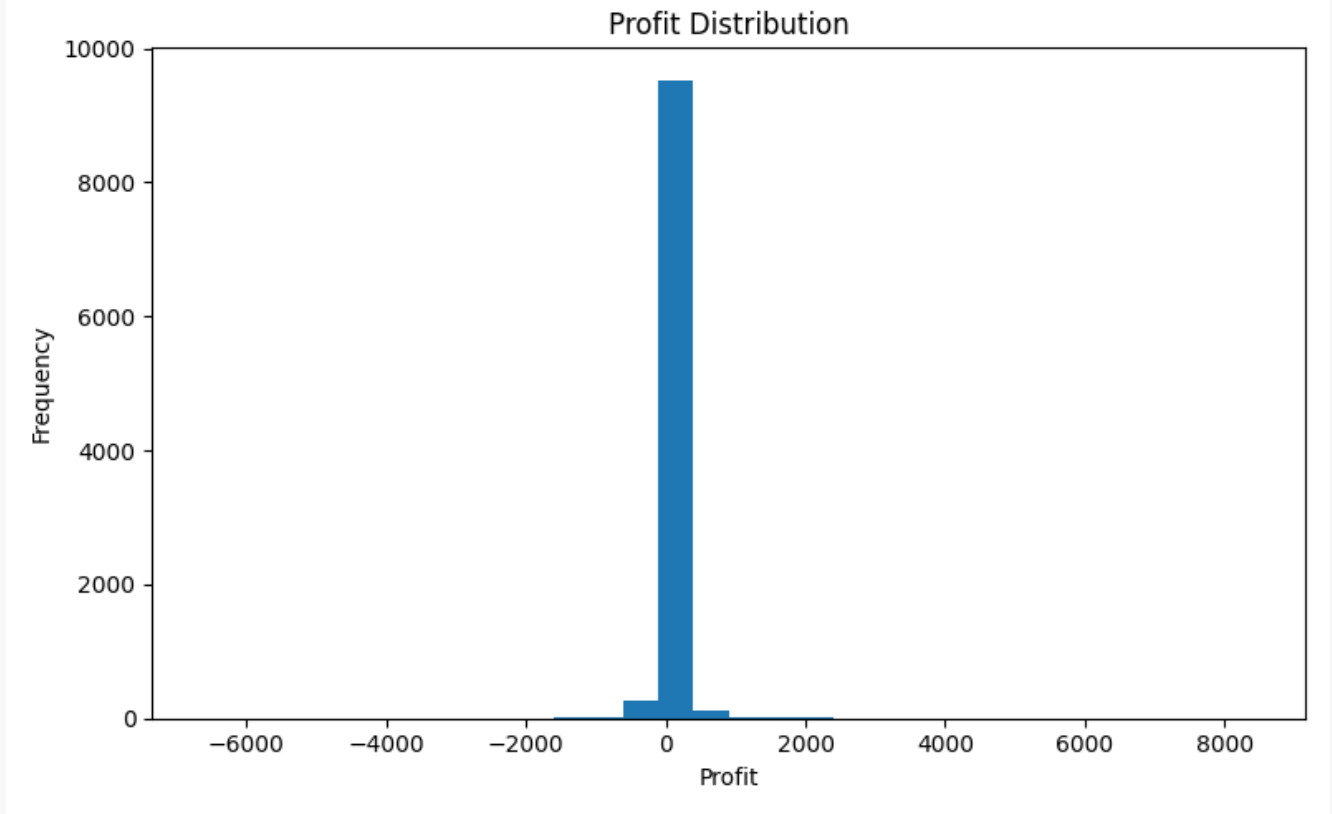

Figure 2. The figure displays the distribution of profit values across customer transactions. Most orders generated moderate profits, while some transactions produced large profits or heavy losses.

Interpretation: Technology products generated the highest overall profit among all categories, indicating strong business performance and better profit margins. Furniture products showed less consistent profitability despite high sales volume. The histogram also revealed several negative-profit transactions, suggesting that discounts or operational costs affected certain orders. Understanding category-wise profitability can help businesses optimize product strategies and pricing decisions.

7.3 Research Question 3

I will investigate if there is a relationship between sales and profit?

A correlation analysis and scatter plot were used to examine the relationship between sales and profit. The Pearson correlation coefficient was calculated using Pandas to measure the strength of the relationship between the two variables.

The analysis showed a positive relationship between higher sales and higher profit.

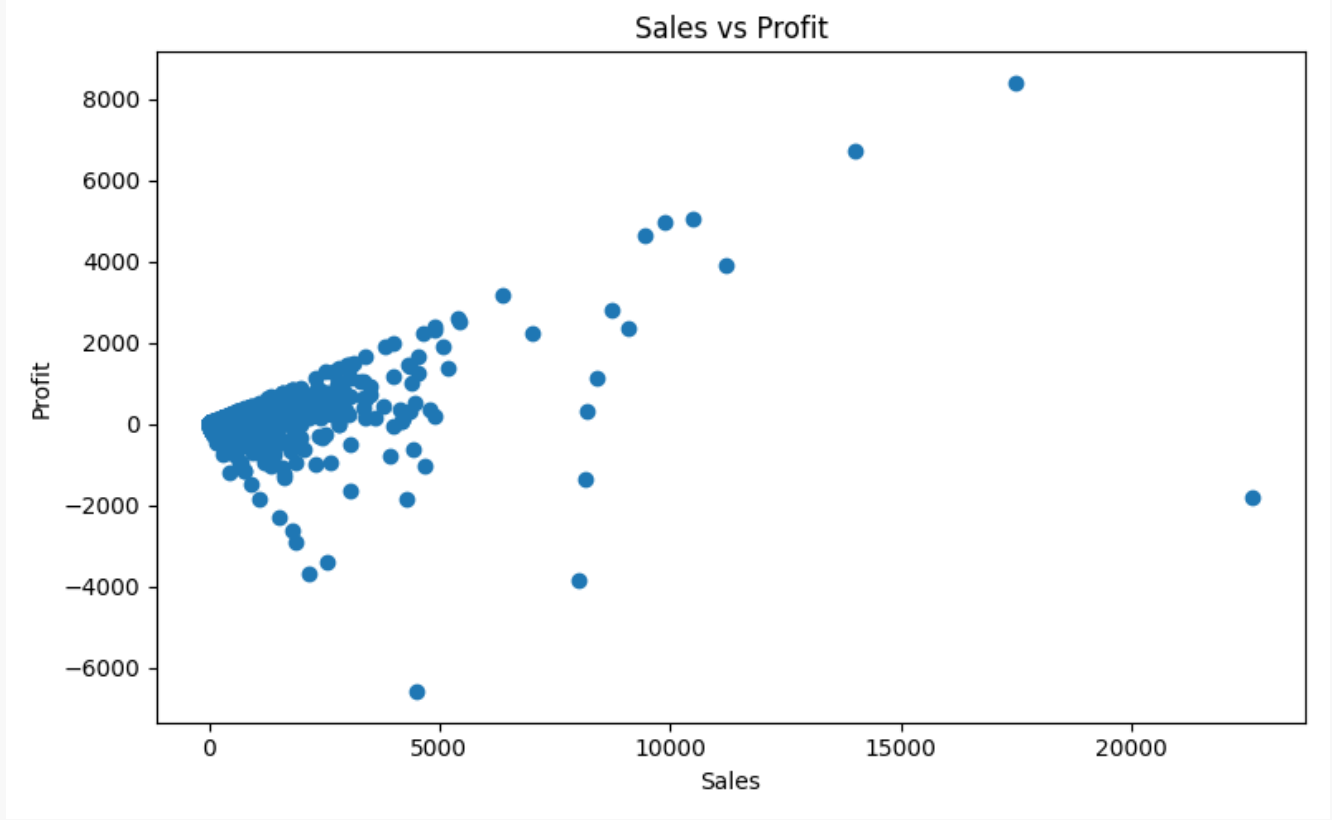

Figure 3. The scatter plot illustrates the relationship between sales and profit for all customer orders. Each point represents an individual transaction.

Interpretation: The scatter plot indicates that higher sales values generally produce higher profits. However, the relationship is not perfectly linear because some high-sales orders resulted in low or even negative profits. This suggests that factors such as discounts and operational expenses may reduce profitability. The correlation analysis confirms a moderate positive relationship between sales and profit overall.

7.4 Additional Visualization

Monthly Sales Trend Analysis

Monthly sales trends were analyzed by extracting month values from the order date column and grouping sales data accordingly.

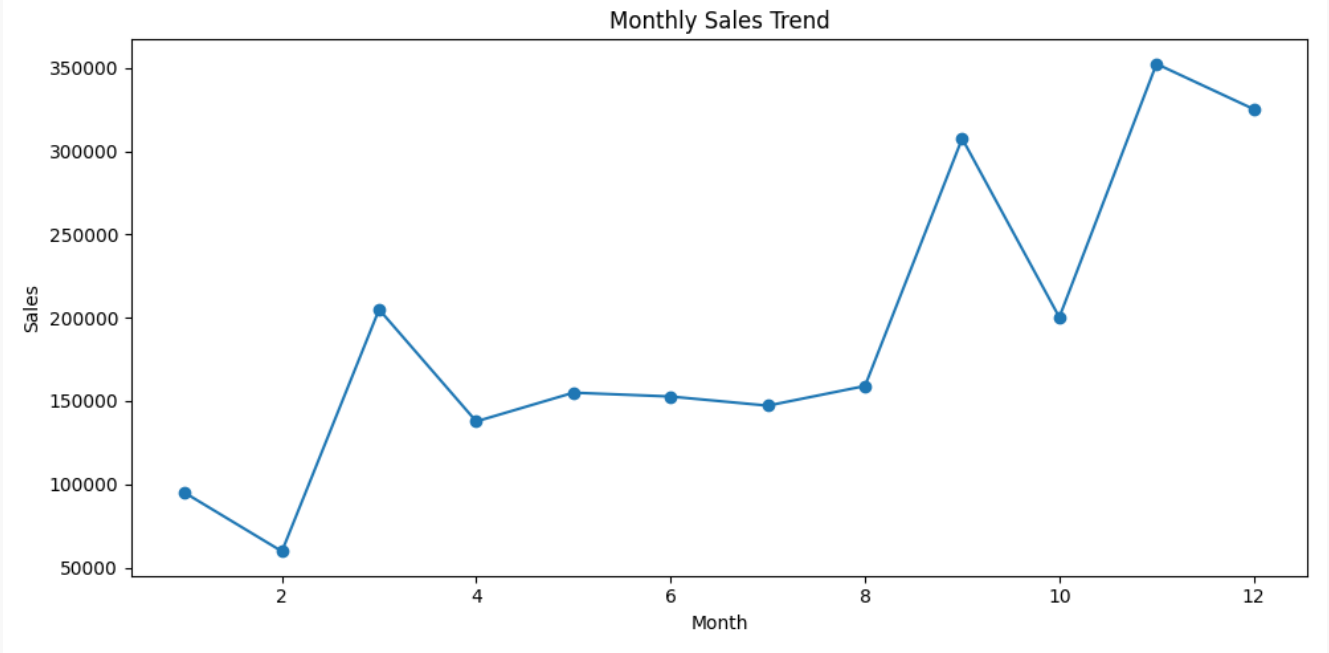

Figure 4. The figure shows how sales changed across different months throughout the year.

Interpretation: Sales varied significantly across months, indicating possible seasonal purchasing behavior. Certain months experienced higher sales activity, suggesting periods of increased customer demand. The trend analysis helps businesses identify peak sales periods and improve inventory and marketing planning.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Z-score calculation using NumPy
Purpose	Detect unusual high-value sales transactions
Formula or logic	Z = (x – mean) / standard deviation
Result / takeaway	Several sales records were identified as statistical outliers

NumPy was used to calculate z-scores for the sales column. Transactions with z-score values greater than 3 or less than -3 were considered outliers. This computation helped identify unusually large sales transactions that differed significantly from the average sales values.

The project also used np.where() to classify orders into High Sales and Low Sales categories based on sales amount.

9. Key Findings

1. The West region generated the highest overall sales revenue among all regions, indicating stronger customer activity and market performance.
2. Technology products produced the highest total profit, making them the most profitable product category in the dataset.
3. Consumer segment customers contributed the largest share of total sales compared to Corporate and Home Office segments.
4. Several transactions generated negative profits despite high sales values, suggesting that discounts or operational costs reduced profitability.
5. A positive relationship exists between sales and profit, although higher sales do not always guarantee higher profit.
6. Outlier analysis identified several unusually large sales transactions that significantly differed from normal order values.

10. Limitations

Although the dataset provided valuable business insights, several limitations exist in this analysis. The dataset does not include external business factors such as advertising campaigns, customer satisfaction, or economic conditions that may influence sales performance.

Some missing values and outlier transactions may slightly affect analytical accuracy. In addition, correlation analysis cannot prove causation, meaning that higher sales do not necessarily cause higher profit. The dataset also represents a limited business scenario and may not fully reflect all retail industries or markets.

11. Conclusion

This project successfully analyzed retail sales and profit data using Pandas, NumPy, and Matplotlib. The analysis explored regional sales performance, category profitability, customer behavior, sales trends, and the relationship between sales and profit.

The findings showed that the West region and Technology category performed best overall. NumPy-based outlier analysis identified unusual sales patterns, while visualizations helped communicate important business insights clearly.

Overall, the project demonstrated how data analytics techniques can support business decision-making and improve understanding of sales performance.

12. References / Dataset Source

1. Superstore Dataset – Kaggle
   https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
2. Pandas Documentation
   https://pandas.pydata.org/
3. NumPy Documentation
   https://numpy.org/
4. Matplotlib Documentation
   https://matplotlib.org/

Final Submission Checklist

☑ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.

☑ I attached the project report as a separate file.

☑ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.

☑ My code uses Pandas, NumPy, and Matplotlib meaningfully.

☑ All bracketed placeholder text and helper notes were replaced or deleted before submission.